When AI Is the Target: Real-World Adversarial Attacks and Defenses

One threat, three real cases, three company defenses

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THREAT AND IMPACT

The threat: tricking AI into the wrong decision

Evasion

Altered input fools a model

Poisoning

Corrupted training data or model

Prompt injection

Hidden instructions hijack an AI assistant

Deepfakes

Faked faces and voices fool people

US$25 million

lost in one deepfake video call (Hong Kong, 2024)

US$40 billion

projected US fraud losses from generative AI by 2027

+700%

deepfake incidents in fintech in 2023

- **Used most:** deepfake fraud against people, and prompt injection against AI assistants, the attack Google, Microsoft and Meta now build defenses for.
- **Most exposed:** autonomous vehicles, finance and, wherever models make clinical calls, healthcare.

Sources: Lalchand et al. (2024); Google GenAI Security Team (2025); Parisi (2019, Ch. 10).

STEP 1 · CASE STUDIES

Three real attacks that made headlines

AUTONOMOUS VEHICLES · EVASION

Tesla: tape on a speed sign

How: About 5 cm of tape made the MobilEye camera read 35 mph as 85.

Impact: The car began accelerating toward 85 mph.

Outcome: A 2020 camera seemed immune; vulnerable ones remain in many Teslas.

FINANCE · DEEPFAKE

Hong Kong: a fake CFO on video

How: Every other person on the video call was a deepfake.

Impact: US$25 million sent to fraudsters.

Lesson: A call-back check would have stopped it.

ENTERPRISE AI · PROMPT INJECTION

Microsoft 365 Copilot

How: Injected instructions could make Copilot disclose data, with no user interaction.

Impact: Rated **9.3 Critical**.

Stopped: Microsoft fixed it server-side; no exploitation.

WHY THEY MADE HEADLINES

None needed a break-in: each fooled a trusted AI, or a person, with crafted input.

Sources: Povolny & Trivedi (2020); Lalchand et al. (2024); Microsoft Security Response Center (2025).

STEP 2 · CORPORATE DEFENSES

How Microsoft, Google and Meta defend their AI

MICROSOFT · RED TEAMING

Attack it before attackers do

Red-teamed **100+** generative AI products.

Humans stay crucial; the work is never finished.

GOOGLE · ADVERSARIAL TRAINING

Layered defense for Gemini

Training on adversarial data “significantly enhanced” its prompt-injection defenses.

Plus classifiers and user confirmation.

META · OPEN GUARDRAILS

LlamaFirewall for AI agents

PromptGuard 2 detects jailbreaks; **AlignmentCheck** audits agent reasoning.

CodeShield blocks insecure code.

HOW THEY TEST

Red teams and automated attack tools probe models before release, then again after every change.

WHAT IS PROVING MOST EFFECTIVE

Layers, not one fix. Microsoft stopped the Copilot flaw before exploitation; deepfake payments need a human check.

Sources: Bullwinkel et al. (2025); Google GenAI Security Team (2025); Chennabasappa et al. (2025).

COMPARISON CHART

Industry comparison: attack types vs. defense strategies

Attack type	Industry most hit	Adversarial training	Red teaming & testing	Input/output classifiers	Anomaly detection	Human verification
Evasion	Autonomous vehicles	○	●	○	●	—
Poisoning	Any model that retrains	○	●	—	●	○
Prompt injection	Enterprise AI assistants	●	●	●	○	●
Deepfakes	Finance	—	○	○	○	●

● strong fit ○ partial fit — little effect *Author's assessment from the cases and defenses on slides 3–4.*

Takeaway: in this assessment, red teaming and anomaly detection help against every attack type; deepfakes need process checks.

Sources: Google GenAI Security Team (2025); Bullwinkel et al. (2025); Chennabasappa et al. (2025); Parisi (2019, Ch. 5).

Applying Chapter 5, and the governance trend that requires it

CHAPTER 5: ANOMALY DETECTION AS A DEFENSE

Learn what normal looks like; flag what does not fit.

- **Inputs:** 85 mph on a 35 mph road is out of range → slow down.
- **Payments:** a large, urgent transfer after one video call → hold and call back.
- **Training data:** screen outliers before retraining → blocks poisoning.
- **AI agents:** an assistant suddenly sending data outside → block it.

TREND: AI GOVERNANCE MAKES ROBUSTNESS LAW

EU AI Act, Article 15: high-risk AI must be "resilient against attempts by unauthorised third parties to alter their use, outputs or performance."

It names data poisoning, model poisoning and adversarial examples, and requires measures to "prevent, detect, respond to" them.

Effect: robustness testing and anomaly detection become legal duties.

Sources: Parisi (2019, Ch. 5); Regulation (EU) 2024/1689 (2024, Art. 15).

Reflection and references

Reflection. In Modules 3–5 I built an anomaly detector, a fraud model and a phishing filter and judged them by accuracy; this module showed the detector itself is a target. I want to build AI defenses that are red-teamed before release, watched by anomaly checks, and that let a human confirm high-stakes actions.

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Chapter citations: Parisi (2019), Chapter 5 “Network Anomaly Detection with AI” and Chapter 10 “Assessing your AI Arsenal.”